



Bhabha University Bhopal (M.P.)

Scheme of Examination

Programme – Master of Technology

Branch – Civil Engineering

Specialization – Construction Technology & Management

Second Semester

Session 2018-19

Sr .No.	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Marks for End Sem. Examination		Total Marks
			L	T	P		End Sem. Exam.	Tests (Two)	Assignments /Quiz	End Sem. Practical /Viva	Practical Record/Assignment/Quiz /Presentation	
1	CE-201	Construction Mgmt.	3	1	-	4	70	20	10	-	-	100
2	CE-202	Prefabrication design & its construction tech.	3	1	-	4	70	20	10	-	-	100
3	CE-203	Construction Equipment and Material Management	3	1	-	4	70	20	10	-	-	100
4	CE-204	Financial Mgt. in Construction Industries	3	1	-	4	70	20	10	-	-	100
5	CE-205	Appropriate Technology and Energy Conservation	3	1	-	4	70	20	10	-	-	100
6	CE-206	Lab-III Cont. mgt. software Lab	-	-	6	6	-	-	-	90	60	150
7	CE-207	Lab-IV Ad. Construction - II	-	-	6	6	-	-	-	90	60	150
		Total	15	5	12	32	350	100	50	180	120	800

L: Lecture T: Tutorial P: Practical

End semester exam will be of 3 hours duration



Bhabha University Bhopal (M.P.)

M.TECH (CTM)

Construction Management

Subject Code-CE201

Unit-1

Contract Management - I

Types of Construction contract, Lump sum, Unit rate, cost plus-fee, Cost Plus percentage-fee, Incentive Contracts, Nature of Contract, Contract Documents and Contracting procedures, contract revisions.

Unit-2

Contract Management - II

Technical Specifications, Drawings, Tender Bond, Labour and Material Payment Bonds, Scrutiny of Tenders, acceptance, letter of indent. Important Contract clauses, Terms of Payment, retention acceptance and final payment, maintenance period, Time for Completion, Extension of time, Variation in work and conditions, claims and disputes, liquidated damages, Termination rights and responsibility of client, Architect, Engineer, Contractor, Professional liability.

Unit-3

Tender Management

Advance Techniques of Estimating. Principles of Analysis of rates and Specification, writing for different types of construction industries, capital structure, Theories.

Unit-4

Legal Frame Work of Construction

Constitutional provisions relating to Business and industry, Master Plans, Indian Contract Act. Arbitration act.

Unit-5

Labour Laws and Legislation

Contract labour (RRA) ACT 1970, laws relating to wages, bonus & industrial disputes.

Refernces:-

1. Construction Management –by Daniel w. Halpin & Bolivar A. Senior
2. Construction Management and Planning –by B. Sengupta & h Guha
3. Modern Construction Management –by Frank Harris and Ronald McCaffer
4. Principles of Construction Management –by Roy Pilcher
5. Construction Management –by Paul Watson



Bhabha University Bhopal (M.P.)

M.TECH (CTM)

Prefabrication design & its construction technology

Subject Code-CE202

Unit-1

Prefabricated Construction

Prefabricated construction, necessity, Advantages, disadvantages, Mass produced steel, reinforced concrete and masonry systems, Industrialised buildings.

Unit-2

Modular Construction

Modular coordination, basic module, planning and design modules, Modular grid systems, National Building Code Specification, Standardisation, Dimensioning of products, Preferred dimensions and sizes, tolerances and deviations layout and processes.

Unit-3

Prefabricates

Classification, foundation, columns, beams, roof and floor panels, wall panels, clay units, box prefabricates, erection and assembly.

Unit-4

Design of prefabricated Elements

Lift points, beams, slabs, columns, wall panels, footings, design of joints to transfer axial forces, moments and shear forces.

Unit-5

Construction Techniques

Large panel construction, Lift slab system, Glover system, constains' jack-block system, Constain V-Plate system, Bis on system, Silber-Kuhi System, control of construction processes. Equipments, horizontal and vertical transportation.

References :-

1. Prefab Architecture –by Ryan E. Smith
2. Prefabrication of Building Facades –by Vitor Abrantes, Barbara Rangel
3. Prefabricated Housing : Construction and Design Manual –by Filipp Moizer and Jutta Albus
4. Prefab Houses –by Marta Serrats
5. Inside Prefab : The Ready Made Interior –by Deborah Schneiderman



Bhabha University Bhopal (M.P.)

M.TECH (CTM)

Construction Equipment and Material Management

Subject Code-CE203

Unit-1

Planning and Selection of Construction Equipment

Advantage of mechanization of Construction industry. Merits of Labour intensive construction. Planning for construction equipments. Analytical studies, equipment operation. Selection of construction machinery & equipments.

Unit-2

Production Estimates, Sizing and Matching

Cycle time capacity ratings and output of Excavators, Power shovels, drag lines, scrapper, bulldozers, tractor shovels rippers, motor graders etc. Sizing and matching. Capacity ratings and output of compactors, aggregate processing plant concrete production plants etc.

Unit-3

Economics of Construction Equipment

Equipment working rates, Investment cost, Depreciation cost, major repair cost. Cost of fuel and lubricants. Cost of labour, servicing and field repairs, overheads. Recommendations of statutory bodies.

Unit-4

System Approach

Problems of equipment management. Application of CPM in equipment management. Application of the assignment model, transportation model and waiting line models in equipment management.

Unit-5

Material Management

Materials planning and budgeting. Role and functions at different levels of management and budgeting variations. Stages of materials management. A.B.C. analysis. Advantages, mechanics purpose cautions, limitations and tabular analysis. Purchasing parameters and inter relationships. Time source quantity, price quality, grading systems. Special purchasing systems. Obsolescence. Scrap disposal.



Bhabha University Bhopal (M.P.)

M.TECH (CTM)

References:-

1. Construction Project Management –by Kumar Neeraj Jha
2. Construction Planning and Management –by Dr. UK Shrivastava
3. Construction Materials, Methods and Techniques –by William P Spence.
4. Construction Methods and Measurement –by S.W. Nunnally
5. Construction Planning Equipment and Methods –by R. Peurifoy



Bhabha University Bhopal (M.P.)

M.TECH (CTM)

Financial Management in Construction Industries

Subject Code-CE204

Unit-1

Personnel Management

Principle of personnel management. Qualities of a personnel manager. Objective of personnel management. Personnel policies, procedures and programmes. Organizational structure of personnel department. Man power resources. Human resource planning. Job analysis. Performance standards, work rules. Recruitment and selection process. Tests and interview. Induction, orientation and indoctrination. Policies, promotion, demotion, transfers etc. Training of personnel. Need for training. Principles of training programmes. Types of training programmes on the job, training policy and implementation. Task analysis, identification and methodologies.

Unit-2

Labour Management

Industrial relations in construction industry. Principles of industrial relationships. Functional requirements and programme, Industrial disputes, causes of disputes. Types of disputes. Procedures of the settlement of industrial disputes. Implementation mechanism. Trade Unions - Principles of industrial trade unionism. Objectives and functions. Essentials of trade union.

Unit-3

Waste Management

Introduction to waste and waste management, the concept of productivity and its inter relationship with productivity. Systems concept of waste. Complementarity of waste and resource management. Identification of construction waste, material waste, man power waste, energy waste, space waste, time waste, equipment waste, capital waste, utilities and services waste.

Unit-4

Financial Management

Managerial Economics & Financial Statement Nature and scope of managerial economics. Economic theories. Demand analysis and forecasting. Elasticities of demand. Cost and production analysis. Pricing decisions, Policies and practices. Break even analysis. Time value of money, Economics. Comparisons using time value of money, basis of comparisons. Decision making amongst alternatives. Cash flow, discounted cash flow. Cash flow forecasting, Project appraisal through financial statements.



Bhabha University Bhopal (M.P.)

M.TECH (CTM)

Unit-5

Capital Generation & Financial Accounting Banking

Financial Institutes like IFCI, IBI, International financing etc. Book keeping process in construction. The accybtacy cycle. Journals, ledgers etc. for labour cost, materials and purchases miscellaneous ledgers.

References :-

1. Construction Accounting & Financial Management – by Steven J. Peterson
2. Financial Management and Accounting for the Construction Industry Vol. 1 – by Matthew Bender
3. Financial Management and Accounting for the Construction Industry Vol. 2 – by Matthew Bender
4. Construction Project Management –by Paul Netscher
5. Financial Management in Construction Contracting –by Andrew Ross & Peter Williams



Bhabha University Bhopal (M.P.)

M.TECH (CTM)

Appropriate Technology and Energy Conservation

Subject Code-CE205

Unit-1

Appropriate technology

concept and its role in the present circumstances.

Unit-2

Rural Housing & Rural Environmental Technologies

Planning, use of locally available materials, construction techniques. Concept and scope in rural areas planning of water supply schemes in rural areas, development of preferred sources of water, springs, wells, infiltration wells infiltration galleries, collection of rain water, specific problems and method's in rural water supply and treatment. Treatment and disposal of waste water, community and sanitary latrines. Compact and simple waste water disposal systems, biogas plants.

Unit-3

Rural roads

Planning of rural roads, Socio-economic aspects, materials for rural roads, design aspects, drainage problems, and maintenance of rural roads.

Unit-4

Energy Conservation

Energy production, distribution and utilization, a review of global situation. Energy Trends, renewable and non renewable sources, research reviews. Building designs and energy factors affecting energy budget in buildings and settlements, Design of buildings for minimising energy. Solar, Wind and Tidal energies, a review and their adoptability.

Unit-5

Low energy materials, construction techniques and environmental control

References-

1. Introduction to Energy Technologies for Efficient Power Generation – Alexander V. Dimitrov
2. Energy Efficiency –by Ming Yang & Xin Yu
3. Sustainable Energy technologies –by Eduardo Rincon Mejia and Alejandro de las Heras
4. Appropriate Technology for Rural Deveiopment –by Derek Miles
5. Towards Global Actionb For Appropriate Technology –by A.S. Bhalla

Bhabha University Bhopal (M.P.)

ADVANCE CONSTRUCTION LAB-II (CE 207)
List of Experiment
1. Aggregate Crushing Value Test
2. Determination of aggregate impact value
3. Determination of Los Angeles Abrasion value
4. Determination of California Bearing Ratio values
5. Determination of penetration value of Bitumen
6. Determination of ductility of the bitumen
7. Determination of flash point and fire point of bituminous material
8. Determination of Bitumen content by centrifuge extractor
9. Determination of stripping value of road aggregate
10. Determination of Marshall stability value for Bituminous mix
11. Determination of shape tests on aggregate

Bhabha University Bhopal (M.P.)